

Supporting Information

Investigating errors in alchemical free energy predictions using random forest models and GaMD

Skanda Sastry^{1†}, Michael Tae-jong Kim¹

¹Protein Analytical Chemistry Department, Genentech Inc, South San Francisco, CA 94080

[†]Corresponding author. Email: sastry.skanda@gene.com

S1 Empirical binding affinity data for each point mutation

System	Point Mutation	Data Source	Reported Kd value	Experimental technique	Converted ddG (kcal/mol)
Kelley hu4D5-5	wild-type hu4D5-5	Kelley 1993, Kelley 1992	0.27, 0.43	SPR, radioimmunoassay	0
Kelley hu4D5-5	VH-W95A	Kelley 1993	5000, 1400	SPR, titration calorimetry	5.07
Kelley hu4D5-5	VL-F53N	Kelley 1993	2.02	SPR	1.193
Kelley hu4D5-5	VH-R50A	Kelley 1993	610	SPR	4.577
Kelley hu4D5-5	VH-Y100aA	Kelley 1993	3400, 340	SPR, titration calorimetry	4.231
Kelley hu4D5-5	VL-T31A	Kelley 1993	1.04	SPR	0.799
Kelley hu4D5-5	VL-E55Y	Kelley 1993, Kelley 1992	0.20, 0.16, reported kcal in 1992	SPR, radioimmunoassay	-0.178
Kelley hu4D5-5	VL-R66G	Kelley 1993, Kelley 1992	0.39, 1.10, reported kcal in 1992	SPR, radioimmunoassay	0.218
Kelley hu4D5-5	VH-A78L	Kelley 1993, Kelley 1992	Reported value in kcal/mol	Radioimmunoassay	0
Kelley hu4D5-5	VH-V102Y	Kelley 1993, Kelley 1992	Reported value in kcal/mol	Radioimmunoassay	0
Kelley hu4D5-5	VL-H91A	Kelley 1993	55	SPR	3.151
Kelley hu4D5-5	VL-T93A	Kelley 1993	1.10	SPR	0.833
Kelley hu4D5-5	VH-F100A	Kelley 1993	1.9	SPR	1.157
Kelley hu4D5-5	VL-Y92A	Kelley 1993	2.70	SPR	1.365
mab1	VH-G to S mutation	Gao 2021	10-fold increase in binding	SPR	-1.365
mab1	VH-G to A mutation	Gao 2021	Complete loss of binding affinity	SPR	0
mab1	VL Y to A mutation	Gao 2021	>10-fold decrease in binding affinity	SPR	>1.365
mab1	VH N to Q mutation	Gao 2021	5-fold increase in binding affinity	SPR	-0.954
mab2	VH G to E mutation	Gao 2021	Similar binding affinity	SPR	0
mab2	VL Y to R mutation	Gao 2021	>4-fold increase in binding affinity	SPR	< -0.82
mab2	VH T to Y mutation	Gao 2021	Similar binding affinity	SPR	0
mab3	VL-Met oxidation	Gao 2021	Similar potency	cell-based assay	0
mab3	VH-Met oxidation	Gao 2021	Similar potency	cell-based assay	0
mab3	VH-Met oxidation #2	Gao 2021	Similar potency	cell-based assay	0

Table S1: Empirical binding affinity data. Kelley 1993 refers to Kelley & O'Connell,¹ Kelley 1992 refers to Kelley et al.,² and Gao 2021 refers to Gao et al.³

Table S1 contains the complete set of empirical mutagenesis data used in this study. For the hu4D5-5 experimental data, if there were multiple published results we defaulted to the measurement using surface plasmon resonance (SPR), but in cases when the dissociation rate was too fast (e.g. W95A and Y100A) we used isothermal titration calorimetry (ITC) based on conversations with Robert Kelley. For mab1/mab2, there were a few qualitative measurements which were not included in the final calculation of RMSE but were still considered in our error cases such as mab2 VL-Y→R.

S2 hu4D5-5 VH-R50A and mab2 VL-Y→R van der Waals step corrections

Case	Original ddG (kcal/mol)	Corrected ddG (kcal/mol)	Empirical ddG (kcal/mol)	Original Error (kcal/mol)	Corrected Error (kcal/mol)	Model R ²	Notes
R50A vdw step	2.98 ± 1.02	3.10 ± 1.02	4.58	1.60	1.48	0.59	Correction based on VL-E71 chi1 rotamer
Y→R vdw step	0.61 ± 1.04	0.44 ± 0.96	< -0.83	>1.43	> 1.27	0.64	Correction based on VL-Y49 chi1/chi2 rotamers

Table S2: van der Waals step corrections for hu4D5-5 VH-R50A and mab2 VL-Y→R. The corrections made trivial change in the error unlike the charging step corrections. Uncertainties are 95% confidence intervals determined from bootstrapping.

As mentioned in the results section, applying the RF + GaMD correction method made trivial impact on the estimated van der Waals step $\Delta\Delta G$ for VH-R50A and mab2 VL-Y→R, two of the most erroneous charge-changing perturbations (Table S2).

S3 Complete TI $\Delta\Delta G$ predictions

System	Mutant	Predicted $\Delta\Delta G$ (kcal/mol)	Predicted $\Delta\Delta G$ SE (kcal/mol)	Empirical $\Delta\Delta G$ (kcal/mol)	Quantitative Error (kcal/mol)	Qualitative Error
hu4D5-5	VH-W95A (b)	3.19 ± 0.71	1.41	5.07	1.88	
hu4D5-5	VL-F53N (b)	-0.56 ± 0.57	0.35	1.19	1.75	
mab2	VH-T→Y	1.69 ± 0.75	0.42	0	1.69	
hu4D5-5	VH-R50A (c)	2.98 ± 1.02	1.77	4.58	1.60	
hu4D5-5	VL-E55Y (b, c)	-1.44 ± 1.09	0.86	-0.18	1.26	
mab1	VH-N→Q	0.2 ± 0.68	0.92	-0.95	1.15	
hu4D5-5	VL-T31A	-0.15 ± 0.45	0.26	0.80	0.95	
hu4D5-5	VL-R66G (c)	-0.71 ± 1.08	1.90	0.22	0.93	
hu4D5-5	VH-F100A (b)	1.67 ± 0.51	0.67	1.16	0.51	
mab2	VH-G→E (c)	0.51 ± 1.03	0.46	0	0.51	
mab1	VH-G→S	-1.81 ± 0.47	0.60	-1.36	0.45	
hu4D5-5	VH-A78L	0.43 ± 0.49	0.41	0	0.43	
hu4D5-5	VH-Y100aA (b)	4.55 ± 0.65	1.60	4.23	0.32	
hu4D5-5	VL-H91A	3.36 ± 0.48	0.84	3.15	0.21	
hu4D5-5	VL-T93A	1.01 ± 0.39	0.32	0.83	0.18	
mab3	VH-Met ox site 1	-0.12 ± 0.30	0.61	0	0.12	
mab3	VH-Met ox site 2	0.11 ± 0.27	0.96	0	0.11	
mab3	VL-Met ox	-0.06 ± 0.20	0.31	0	0.06	
hu4D5-5	VH-V102Y (b)	-0.05 ± 0.56	0.65	0	0.05	
hu4D5-5	VL-Y92A (b)	1.35 ± 0.66	1.49	1.36	0.01	
mab1	VH-G→A	0.48 ± 0.35	0.85	“Complete loss of binding”	-	Inaccurate
mab1	VL-Y→A (b)	3.58 ± 0.59	0.84	> 1.36	-	Accurate
mab2	VL-Y→R (b, c)	0.61 ± 1.04	0.95	< -0.82	> 1.44	Inaccurate

Table S3: Complete TI $\Delta\Delta G$ predictions dataset (n = 23). SE = standard error. Quantitative error is $|\text{Predicted } \Delta\Delta G - \text{Empirical } \Delta\Delta G|$. In the Mutant column, (b) means bulky side chain, (c) means charge-changing, (b, c) means both.

Table S3 contains the complete set of TI $\Delta\Delta G$ predictions, including all cases with qualitative empirical data. Note that there are *two* different measures of uncertainty given in this table. The standard errors in the **Predicted $\Delta\Delta G$ SE** column are calculated by taking the standard error of the mean given n=5 biological replicates. Only the error bars in main text Figure 1 use these standard errors.

The uncertainty metrics after the $\pm$ symbol in the **Predicted $\Delta\Delta G$** column are computed by aggregating the data from all 5 replicates, bootstrapping (see final subsection of the Methods section in the main text), and deriving 95% confidence intervals. The bootstrapping 95% confidence interval uncertainties were computed to allow a consistent comparison with the uncertainties of the corrected $\Delta\Delta G$ estimates.

S4 Feature importances and DOF definitions for top 5 most inaccurate cases

S4.1 Feature importances

Rank	hu4D5-5 VH-W95A	hu4D5-5 VH-R50A (charging step)	mab2 VL-Y49R (charging step)	mab2 VH-T53Y	hu4D5-5 VL-F53N
1	0.19	0.15	0.10	0.55	0.24
2	0.18	0.10	0.08	0.11	0.12
3	0.08	0.09	0.07	0.08	0.08
4	0.08	0.09	0.07	0.05	0.08
5	0.04	0.05	0.07	0.03	0.07

Table S4: Feature importance for each of the features given in Table 2 of the main paper

In Table 1 of the main article, the top 5 most influential DOFs given by the random forest model analyzing the TI lambda production data for each of the 5 most erroneous cases was given. In Table S4, the feature importance for each feature included in Table 1 is given. The feature importance metrics were computed using the mean of the built-in Gini impurity decrease in scikit-learn.⁴

S4.2 Definitions of degrees of freedom

The following are definitions of each feature given in the main paper Table 1. CPPTRAJ⁵ was used to compute all geometric degrees of freedom (DOF).

S4.2.1 hu4D5-5 VH-W95A

- VH-W95 chi1: Dihedral angle between C, CA, CB, CG for VH-W95
- VH-W95 chi2: Dihedral angle between CA, CB, CG, and CD1 for VH-W95
- VH-R94 chi4: Dihedral angle between CG, CD, NE, and CZ for VH-R94
- VH-F100 chi1: Dihedral angle between C, CA, CB, CG for VH-F100
- Ag-E71 chi3: Dihedral angle between CB, CG, CD, OE1 for Ag-E71

S4.2.2 hu4D5-5 VH-R50A charging step

- Ag-E71:VH-R50 salt bridge distance: Distance between Ag-E71 CD atom and VH-R50 CZ atom
- VH-R50 chi1: Dihedral angle between C, CA, CB, and CG for VH-R50

- VH-V48 chi1: Dihedral angle between C, CA, CB, and CG1 for VH-V48
- VL-T94:VH-R50 H-Bond distance: Distance between VL-T94 OG1 and VH-R50 CZ
- Ag-E71 chi3: Dihedral angle between CB, CG, CD, OE1 for Ag-E71

S4.2.3 mab2 VL-Y→R charging step

- Ag-D161:VL-R49 salt bridge distance: Distance between Ag-D161 CG atom and VL-R49 CZ atom
- VL-R49:VL-S50 H-bond distance: Distance between VL-R49 CZ atom and VL-S50 OG atom
- VL-S53 chi1: Dihedral angle between C, CA, CB, and OG atoms for VL-S53
- VL-S50 chi1: Dihedral angle between C, CA, CB, and OG atoms for VL-S50
- Ag-R157 chi1: Dihedral angle between C, CA, CB, and CG for Ag-R157

S4.2.4 mab2 VH-T→Y

- Ag-V117:VH-Y53 H-bond distance: Distance between Ag-V117 backbone O atom (carbonyl oxygen) and VH-Y53 side chain hydroxyl H atom
- VL-Y53 chi1: Dihedral angle between C, CA, CB, and CG atoms for VL-Y53
- VL-T53 chi1: Dihedral angle between C, CA, CB, and CG2 atoms for VL-T53
- VL-T53:VL-N51 H-bond distance: Distance between OG1 atom of VL-T53 and CG atom of VL-N51
- VL-Y53 chi2: Dihedral angle between CA, CB, CG, and CD1 of VL-Y53

S4.2.5 hu4D5-5 VL-F53N

- Ag-C117:VL-N53 H-bond distance: Distance between Ag-C117 backbone O atom (carbonyl oxygen) and VL-N53 CG atom
- VL-N53 chi1: Dihedral angle between C, CA, CB, and CG atoms of VL-N53
- Ag-M102 chi1: Dihedral angle between C, CA, CB, and CG atoms of Ag-M102
- VL-N53 chi2: Dihedral angle between CA, CB, CG, and OD1 atoms of VL-N53
- Ag-N120 chi1: Dihedral angle between C, CA, CB, and CG atoms of Ag-N120

S5 hu4D5-5 VH-R50A and mab2 VL-Y→R one-step corrections

All of our charge-changing transformations were done using the two-step approach (charging step and van der Waals step). Recently, novel smoothstep softcore potentials⁶ were published to avoid certain instabilities associated with charge-changing transformations while doing one-step transformations. When we calculated the $\Delta\Delta G$ for hu4D5-5 VH-R50A and mab2 VL-Y→R via a one-step transformation with smoothstep potentials, the predictions were still affected by the disruption of key salt bridges – i.e. smoothstep potentials did not mitigate the observed issue with salt bridge violations (Figure S1, Figure S3). Once these salt bridges were enforced with NMR restraints (Figure S2, Figure S4), the results improved just as in the two-step transformations (Table S5).

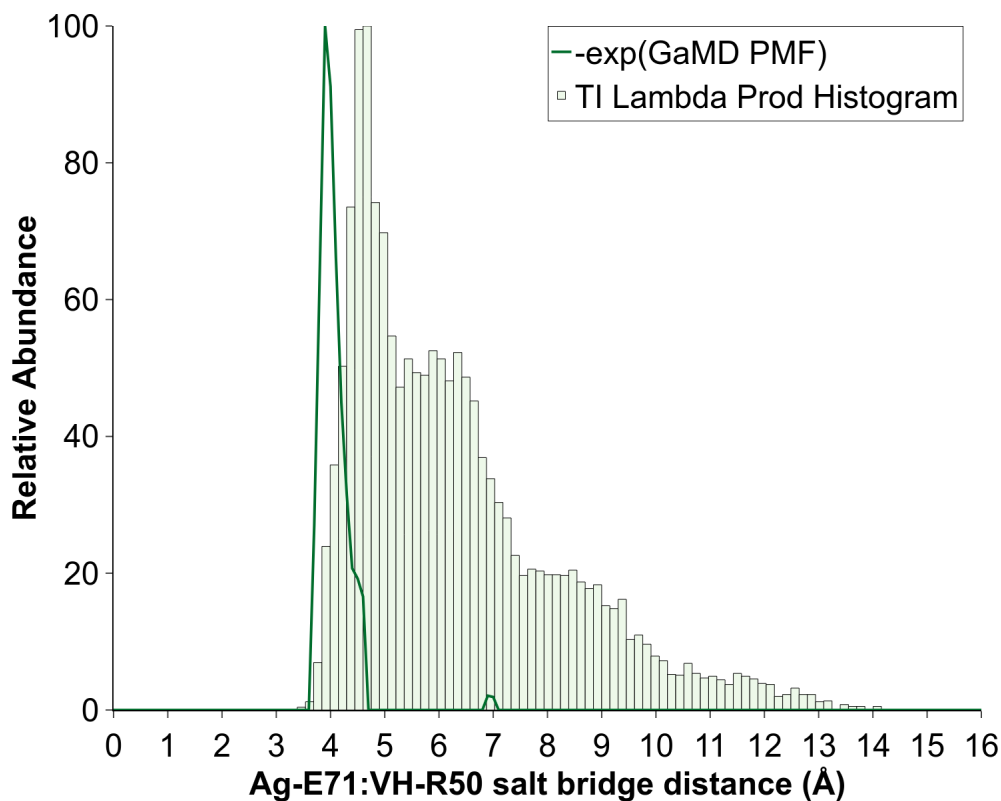


Figure S1: Lambda production visualization of VH-R50. . . Ag-E71 salt bridge during one-step transformation

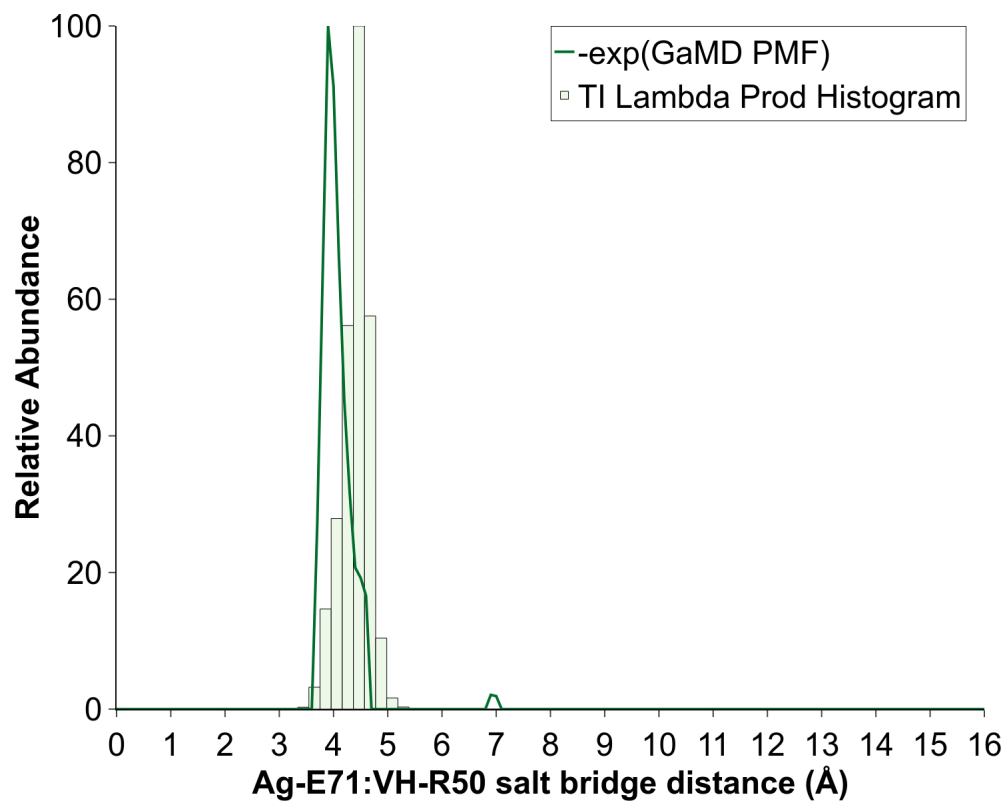


Figure S2: Lambda production visualization of VH-R50 . . . Ag-E71 salt bridge during one-step transformation, with NMR restraints.

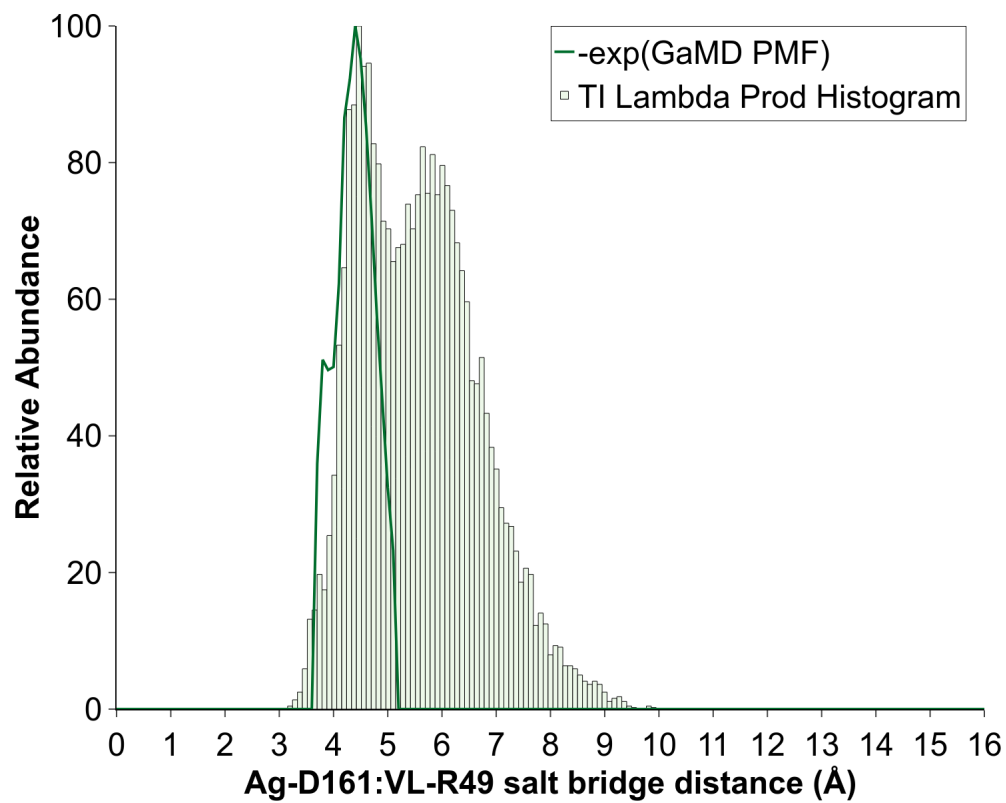


Figure S3: Lambda production visualization of mab2 VL-R49...Ag-D161 salt bridge during one-step transformation

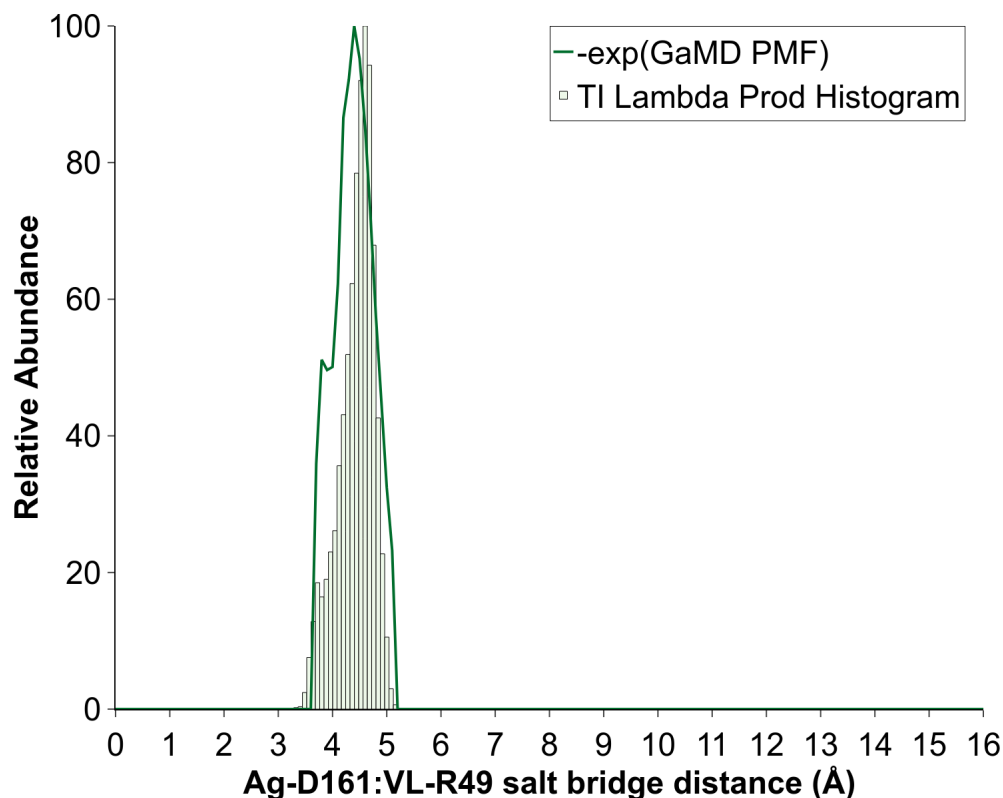


Figure S4: Lambda production visualization of VL-R49... Ag-D161 salt bridge during one-step transformation with NMR restraints

Case	Original ddG (kcal/mol)	Corrected ddG (kcal/mol)	Empirical ddG (kcal/mol)	Original Error (kcal/mol)	Corrected Error (kcal/mol)
R50A one step	2.30 ± 1.22	4.07 ± 1.23	4.58	2.28	0.51
Y→R one step	-0.54 ± 1.31	-3.25 ± 1.16	< -0.83	> 0.29	-

Table S5: Thermodynamic integration results for one-step transformations. Uncertainties are 95% confidence intervals computed via bootstrapping.

S6 Barstar-barnase key distance GaMD 1-D PMFs

In Zhang et al.'s study estimating $\Delta\Delta G$ s for point mutations at the barstar-barnase interface, the mutations Q83R/R83Q, D35A/A35D, R87A, A29Y, and K27A all had mean errors > 2.0 kcal/mol across their four simulation types (10 ns AREX, 10 ns AREST, 50 ns AREX, 50 ns AREST).⁷ We ran GaMD⁸ (5 x 300 ns) and searched for any possible salt bridges or hydrogen bonds involving these amino acids. We share the 1-D

free energy profiles of theorized key distances below.

S6.1 Q83R, R83Q

For Q83R/R83Q, we found a salt bridge between the side chains of barnase residue R83 and barstar residue D39 (Figure S5). We also found a salt bridge between the side chains of barnase residues R83 and D75 (Figure S6).

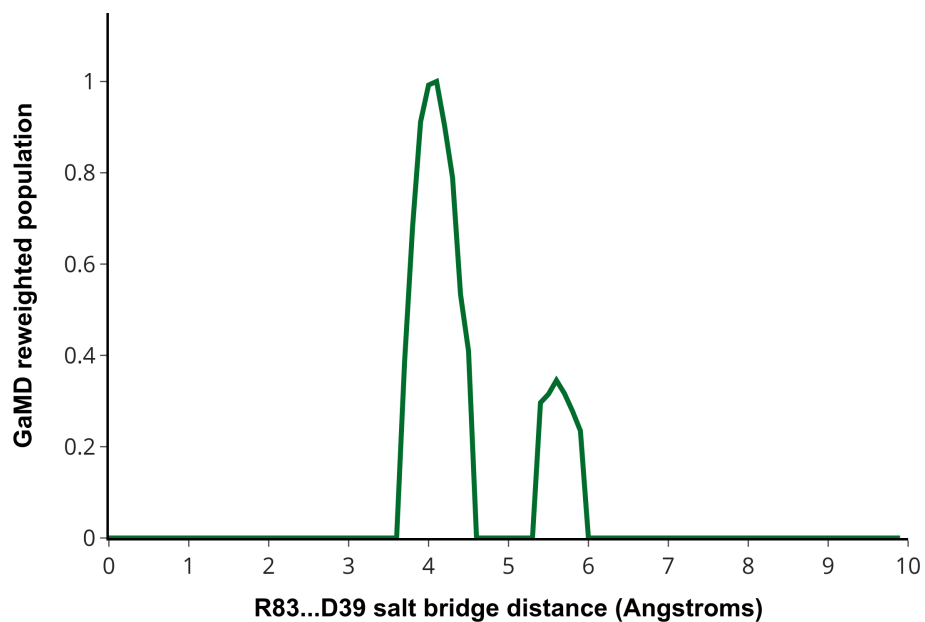


Figure S5: 1-D GaMD free energy profile of Barnase-R83:Barstar-D39 salt bridge

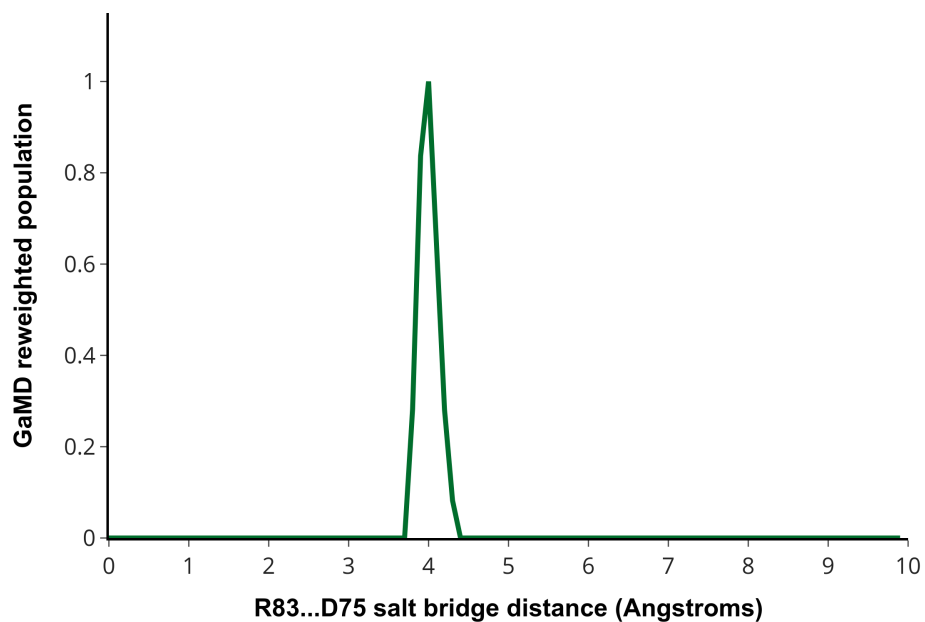


Figure S6: 1-D GaMD free energy profile of Barnase-R83:Barnase-D75 salt bridge

S6.2 D35A, A35D

For D35A/A35D, we found a hydrogen bond between barstar residue D35 side chain and barnase residue R59 backbone (Figure S7).

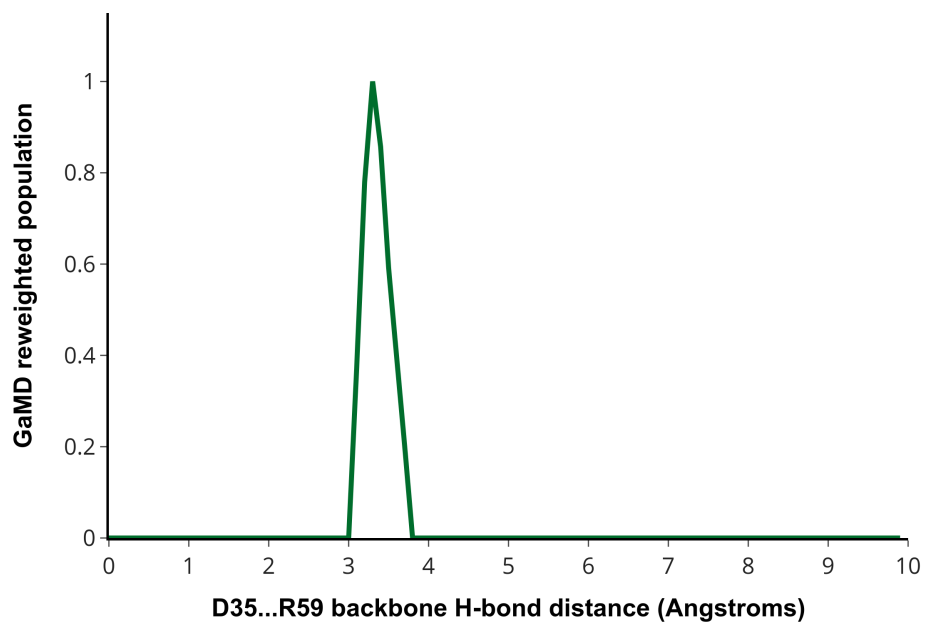


Figure S7: 1-D GaMD free energy profile of Barstar-D35:Barnase-R59 backbone H-bond

S6.3 R87A

For R87A, we found a salt bridge between the side chains of barnase residue R87 and barstar residue D39 (Figure S8).

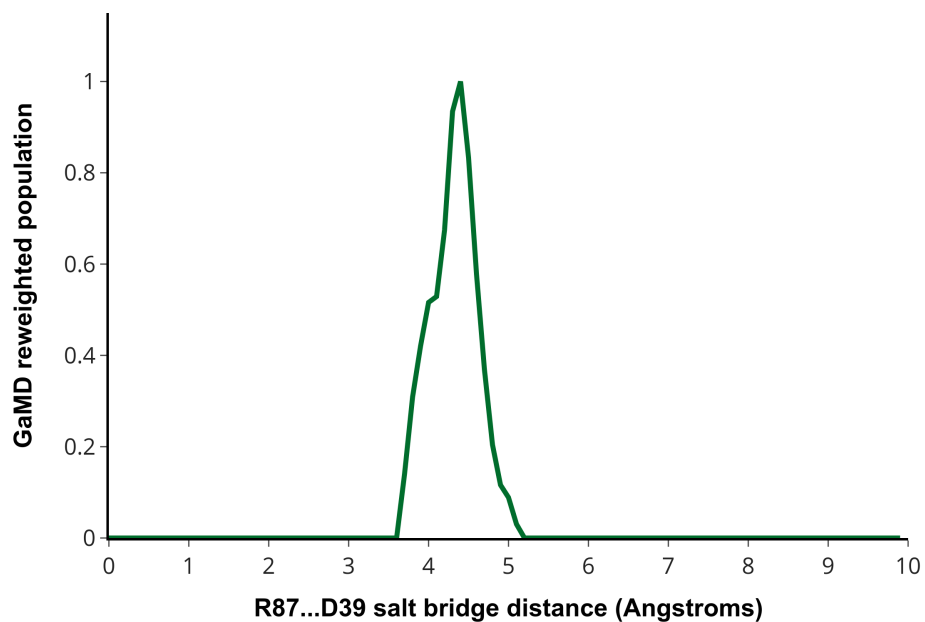


Figure S8: 1-D GaMD free energy profile of Barnase-R87:Barstar-D39 salt bridge

S6.4 A29Y

For A29Y, we found a hydrogen bond between the side chain hydroxyl on barstar residue Y29 and the backbone of barnase residue R83 (Figure S9).

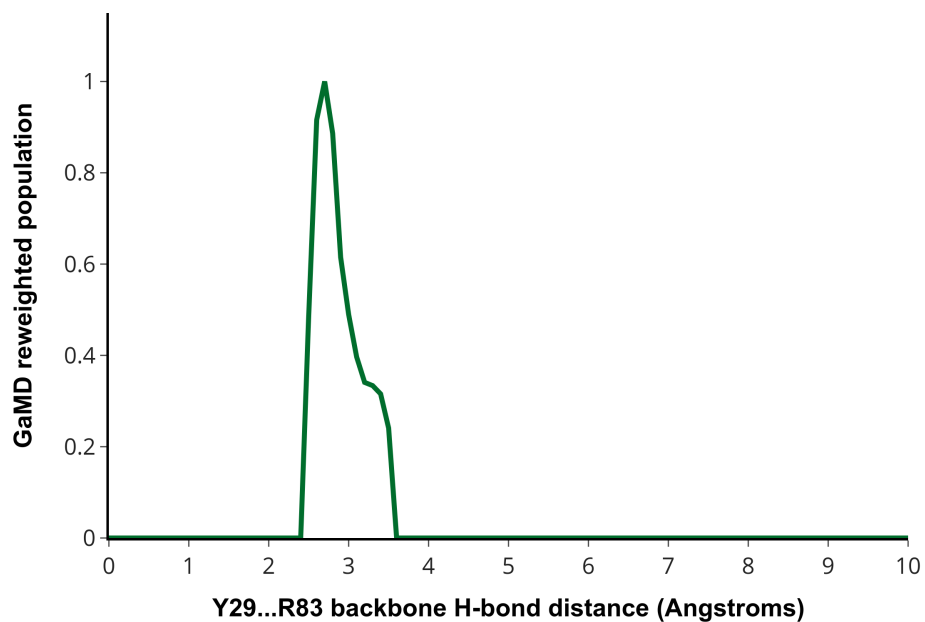


Figure S9: 1-D GaMD free energy profile of Barstar-Y29:Barnase-R83 backbone H-bond

S6.5 K27A

For K27A, we found a salt bridge between the side chains of barnase residue K27 and barstar residue D39 (Figure S10). We also found a salt bridge between the side chains of barnase residue K27 and barnase residue D54 (Figure S11).

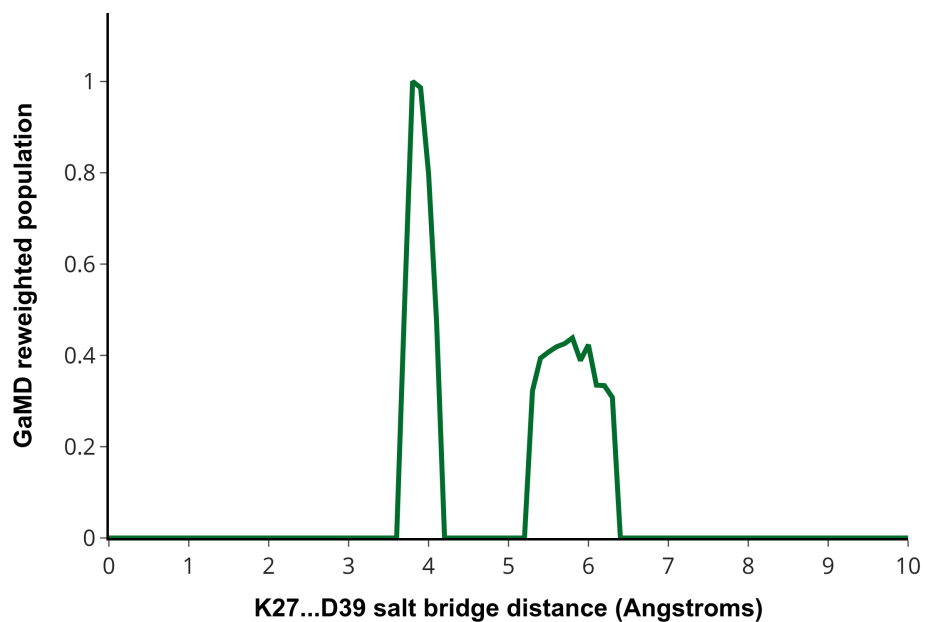


Figure S10: 1-D GaMD free energy profile of Barnase-K27:Barstar-D39 salt bridge

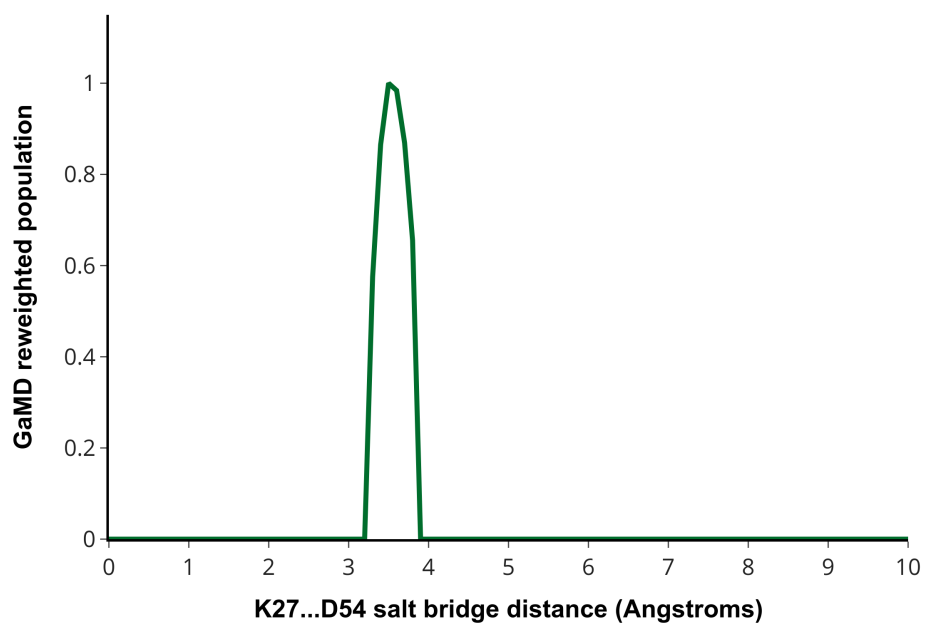


Figure S11: 1-D GaMD free energy profile of Barnase-K27:Barnase-D54 salt bridge

S7 Applying random forest model to all hu4D5-5 cases

System	Original Prediction (kcal/mol)	Standard error	Empirical value (kcal/mol)	Model R^2	Corrected prediction (kcal/mol)
VH-W95A	3.19 ± 0.71	1.41	5.07	0.53	4.63 ± 0.47
VL-F53N	-0.56 ± 0.57	0.35	1.19	0.64	-0.56 ± 0.57
VH-R50A	2.98 ± 1.02	1.77	4.58	0.73 (crg), 0.59 (vdw)	5.14 ± 1.05
VL-E55Y	-1.44 ± 1.09	0.86	-0.18	0.59 (crg), 0.33 (vdw)	-0.81 ± 1.08
VL-T31A	-0.15 ± 0.45	0.26	0.80	0.43	0.24 ± 0.36
VL-R66G	-0.71 ± 1.08	1.90	0.22	0.66 (crg), 0.53 (vdw)	-0.22 ± 0.70
VH-F100A	1.67 ± 0.51	0.67	1.16	0.29	1.67 ± 0.51
VH-A78L	0.43 ± 0.49	0.41	0	0.10	0.43 ± 0.49
VH-Y100aA	4.55 ± 0.65	1.60	4.23	0.59	4.25 ± 0.54
VL-H91A	3.36 ± 0.48	0.84	3.15	0.34	3.36 ± 0.48
VL-T93A	1.01 ± 0.39	0.32	0.83	0.34	1.01 ± 0.39
VH-V102Y	-0.05 ± 0.56	0.65	0	0.34	-0.05 ± 0.56
VL-Y92A	1.35 ± 0.66	1.49	1.36	0.64	1.81 ± 0.55

Table S6: Results when applying error analysis/correction approach to all 13 hu4D5-5 cases

Although the model was designed to investigate the source of error in our most erroneous $\Delta\Delta G$ predictions, we also applied our model to all hu4D5-5 cases ($n = 13$) to observe whether the correction approach increased the error in any predictions (Table S6). We confirmed that the approach indeed decreased the prediction error in all but 1 case (VL-Y92A, in which error increased by 0.44 kcal/mol). The original prediction MAE for the Kelley hu4D5-5 cases was 0.82 ± 0.18 kcal/mol, and after corrections the MAE decreased to 0.53 ± 0.16 kcal/mol. The RMSE similarly decreased from 1.06 ± 0.22 kcal/mol to 0.70 ± 0.18 kcal/mol.

S8 Results of TI protocol with extended 25 ns lambda production

System	Mutant	Predicted $\Delta\Delta G \pm CI$ (kcal/mol)	Predicted $\Delta\Delta G$ SE (kcal/mol)	Empirical $\Delta\Delta G$ (kcal/mol)	Quantitative Error (kcal/mol)	Qualitative Error
hu4D5-5	VL-R66G (c)	-1.59 ± 1.09	1.46	0.22	1.81	
hu4D5-5	VH-W95A (b)	3.63 ± 0.68	1.05	5.07	1.44	
mab2	VH T→Y (b)	1.28 ± 0.70	0.63	0	1.28	
hu4D5-5	VL-T31A	-0.47 ± 0.41	0.24	0.80	1.27	
mab1	VH N→Q	0.31 ± 0.69	0.75	-0.95	1.26	
hu4D5-5	VL-E55Y (b, c)	-1.28 ± 1.07	0.69	-0.18	1.10	
hu4D5-5	VL-F53N (b)	0.34 ± 0.58	0.46	1.19	0.85	
hu4D5-5	VH-R50A (c)	3.95 ± 0.98	1.22	4.58	0.63	
mab2	VH G→E (c)	0.63 ± 1.09	0.33	0	0.63	
mab3	VH-Met site 2	-0.48 ± 0.25	0.49	0	0.48	
hu4D5-5	VH-A78L	0.38 ± 0.47	0.29	0	0.38	
mab3	VH-Met ox site 1	0.37 ± 0.28	0.35	0	0.37	
hu4D5-5	VL-Y92A (b)	1.72 ± 0.64	0.71	1.36	0.36	
mab1	VH-G→S	-1.59 ± 0.47	0.56	-1.36	0.23	
hu4D5-5	VH-Y100aA (b)	4.01 ± 0.62	1.19	4.23	0.22	
mab3	VL-Met ox	0.19 ± 0.21	0.17	0	0.19	
hu4D5-5	VH-F100A (b)	1.31 ± 0.52	0.34	1.16	0.15	
hu4D5-5	VL-T93A	0.97 ± 0.39	0.34	0.83	0.14	
hu4D5-5	VL-H91A	3.27 ± 0.51	1.41	3.15	0.12	
hu4D5-5	VH-V102Y (b)	0.01 ± 0.60	0.58	0	0.01	
mab2	VL Y→R (b, c)	0.16 ± 1.03	0.49	<-0.82		Inaccurate
mab1	VL Y→A (b)	3.54 ± 0.59	0.85	>1.36		Accurate
mab1	VH-G→A	1.06 ± 0.36	0.58	"Complete loss of binding"		Inaccurate

Table S7: TI $\Delta\Delta G$ predictions dataset with original protocol extended to 25 ns per λ ($n = 23$). SE = standard error. Quantitative error is $|\text{Predicted } \Delta\Delta G - \text{Empirical } \Delta\Delta G|$. In the Mutant column, (b) means bulky side chain, (c) means charge-changing, (b, c) means both.

The previous section demonstrates that our RF+GaMD correction method improved the RMSE and MAE of 13 single amino acid residue point mutation predictions. To explore whether similar improvements in accuracy could be achieved simply by using our initial protocol but with more sampling, we re-estimated the $\Delta\Delta G$ s of our entire dataset with λ production extended to 25 ns per λ , 5-fold greater than our original 5 ns per λ (Table S7). Restricting to only the hu4D5-5 data to compare to the data in Table S6, the MAE is 0.71 ± 0.18 and the RMSE is 0.93 ± 0.23 - only a slight improvement over the original hu4D5-5 predictions. Of the most erroneous cases in our original set of predictions, VH-R50A and VL-F53N improve to within 1 kcal/mol error, although in the case of VH-R50A the one-step transformation was still highly inaccurate when sampling was extended to 25 ns per λ (Table S8). Many cases, such as VL-R66G, VL-T31A, and mab1 VH-N→Q became less accurate with the extended protocol.

System	Mutant	Predicted $\Delta\Delta G$ (kcal/mol)	Predicted $\Delta\Delta G$ SE (kcal/mol)	Empirical $\Delta\Delta G$ (kcal/mol)	Quantitative Error (kcal/mol)
hu4D5-5	VH-R50A	1.42 ± 1.14	1.22	4.58	3.16
mab2	VL-Y→R	1.25 ± 1.30	0.86	< -0.82	> 2.07

Table S8: Results from one-step protocol, extended to 25 ns for hu4D5-5 VH-R50A and mab2 VL-Y→R

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